

GAROUA, Cameroon

WMO: 648600

Lat: 9.336N

Lon: 13.370E

Elev: 242

StdP: 98.45

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	16.8	17.7	-4.6	2.6	35.2	-2.5	3.1	34.0	6.1	32.0	5.4	32.5	0.6	0	0.364

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	11.6	41.8	20.9	40.9	21.1	39.9	21.2	27.1	33.3	26.7	32.9	26.4	32.6	3.1	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.7	21.6	29.7	25.2	20.9	29.4	24.9	20.6	29.2	87.1	33.3	85.3	33.1	83.8	33.0	30.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.7	6.5	5.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			9.3	28.8	0.7	1.1	8.8	29.6	8.5	30.2	8.1	30.9	7.6	31.7	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	29.2	26.4	29.6	33.3	34.1	31.6	28.8	27.6	27.0	27.2	29.2	29.1	26.3	(6)	(7)
	DBStd	3.15	2.16	2.52	1.77	1.92	2.39	2.12	1.69	1.52	1.53	1.37	1.49	2.04	(7)	(8)
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	(9)
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	(10)
	CDD10.0	6999	507	549	723	723	669	564	547	527	517	594	573	505	(10)	(11)
	CDD18.3	3958	249	316	465	473	411	314	288	268	267	336	323	247	(11)	(12)
	CDH23.3	50175	3013	4418	7376	7383	5778	3695	2922	2402	2506	3804	3999	2879	(12)	(13)
	CDH26.7	28383	1674	2727	5012	5014	3459	1773	1145	832	967	1861	2351	1567	(13)	(14)
(14) Wind	WSAvg	2.3	1.6	1.7	2.4	3.3	3.1	3.0	2.8	2.5	2.2	2.1	1.6	1.6	(14)	(15)
(15) Precipitation	PrecAvg	1014	0	0	2	45	113	140	196	247	201	70	2	1	(15)	(16)
	PrecMax	1497	0	0	31	158	216	337	377	495	338	213	35	39	(16)	(17)
	PrecMin	468	0	0	0	0	4	28	74	90	53	2	0	0	(17)	(18)
	PrecStd	205	0	0	6	40	53	58	69	85	67	50	6	7	(18)	(19)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.1	41.8	42.5	43.0	41.0	37.8	35.2	33.9	34.2	37.3	39.1	38.2	(19)	(20)
		MCWB	19.0	19.4	19.9	21.8	24.6	25.9	26.1	25.9	26.5	22.8	20.5	19.1	(20)	(21)
	2%	DB	37.8	40.2	41.8	41.9	39.3	36.1	33.8	32.8	33.2	36.2	38.0	36.9	(21)	(22)
		MCWB	18.4	19.1	20.2	21.8	25.1	26.0	25.9	25.6	26.2	23.8	20.6	18.8	(22)	(23)
	5%	DB	36.1	39.0	41.0	40.9	38.2	34.9	32.8	31.9	32.6	35.1	37.0	35.7	(23)	(24)
		MCWB	17.8	19.1	19.9	22.5	25.4	25.9	25.7	25.4	25.9	24.8	20.7	18.3	(24)	(25)
	10%	DB	34.2	37.3	40.0	39.8	37.0	33.4	31.7	30.9	31.6	34.0	36.0	34.0	(25)	(26)
		MCWB	17.0	18.3	19.6	23.1	25.3	25.7	25.3	25.2	25.6	25.6	20.8	17.6	(26)	(27)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.7	23.9	25.6	26.8	27.5	27.4	27.1	26.6	27.2	27.6	26.1	23.6	(27)	(28)
		MCDB	34.2	35.6	36.8	38.2	34.9	34.2	32.5	31.2	32.3	32.9	32.9	32.9	(28)	(29)
	2%	WB	20.1	22.4	24.7	26.0	26.7	26.7	26.2	26.1	26.6	27.0	25.3	21.5	(29)	(30)
		MCDB	33.8	34.5	36.0	36.9	34.7	33.1	31.5	30.9	31.6	32.5	32.5	31.7	(30)	(31)
	5%	WB	18.3	20.5	24.1	25.6	26.4	26.3	26.0	25.7	26.2	26.6	24.6	19.2	(31)	(32)
		MCDB	34.7	35.6	35.1	36.0	34.3	32.6	31.2	30.3	31.0	32.0	31.4	32.8	(32)	(33)
	10%	WB	17.4	18.9	23.2	25.1	26.0	26.0	25.6	25.4	25.9	26.2	23.8	18.2	(33)	(34)
		MCDB	33.4	36.5	34.4	35.2	34.0	32.1	30.5	29.7	30.5	31.6	29.9	32.4	(34)	(35)
(35) Mean Daily Temperature Range	5% DB	MDBR	16.5	15.7	13.8	11.6	10.1	8.9	7.8	7.4	8.2	10.1	15.0	16.1	(35)	(36)
		MCDBR	18.1	16.7	14.7	12.6	11.5	10.5	9.1	8.8	9.6	11.5	15.9	17.2	(36)	(37)
	5% WB	MCWBR	6.0	5.8	5.3	4.4	3.3	3.4	3.1	3.0	3.7	3.8	4.5	5.1	(37)	(38)
		MCWBR	17.6	15.0	11.8	11.3	10.3	9.5	8.7	8.2	9.2	9.8	12.3	16.1	(38)	(39)
(39) Clear-Sky Solar Irradiance	taub		0.556	0.681	0.747	0.700	0.595	0.556	0.508	0.491	0.508	0.523	0.480	0.488	(39)	(40)
		taud	1.818	1.570	1.488	1.633	1.861	1.987	2.150	2.210	2.139	2.070	2.091	1.995	(40)	(41)
	Ebn at Noon		760	683	647	672	730	750	791	818	813	797	823	810	(41)	(42)
		Edh at Noon	213	281	309	264	206	179	153	147	159	168	161	175	(42)	(43)
(43) All-Sky Solar Radiation	RadAvg		5.85	6.01	6.29	6.18	5.87	5.44	5.08	4.76	5.25	5.87	6.01	5.78	(43)	(44)
	RadStd		0.17	0.21	0.32	0.28	0.28	0.21	0.25	0.24	0.22	0.17	0.16	0.15	(44)	(45)

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page